USN | | | | | | | | | | [22CSBS116]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Define overfitting and underfitting	L1	CO1	**02**
	b.	Classify Supervised Learning and Unsupervised Learning with examples	L2	CO2	**02**
	c.	Define backpropagation algorithm	L1	CO3	**02**
	d.	List out the two types of clustering algorithms	L1	CO4	**02**
	e.	State anomaly detection	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Illustrate the key components of a machine learning model	L2	CO1	**05**
		OR			
	b.	Classify the types of learning models based on data input	L2	CO2	**05**
3.	**a.**	Compare supervised and unsupervised learning	L4	CO2	**05**
		OR			
	b.	Differentiate between classification and regression problems in machine learning	L4	CO2	**05**
4.	**a.**	Describe the importance of the F1 score in classification problems	L2	CO2	**05**
		OR			
	b.	Compare the least squares regression and multivariable regression	L4	CO2	**05**
5.	**a.**	Explain the working of a simple Naive Bayes classifier	L2	CO3	**05**
		OR			
	b.	Differentiate between bagging and boosting in ensemble methods	L4	CO3	**05**
6.	**a.**	Describe the role of regularization in regression models	L2	CO4	**05**
		OR			
	b.	Classify clustering techniques based on the type of data they handle	L4	CO4	**05**

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7.	**a.**	Explain the concept of part-of-speech tagging in natural language processing	L2	CO5	**05**
		OR			
	b.	Describe the process of mining association rules using the Apriori algorithm	L2	CO5	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Analyze the advantages and limitations of various machine learning algorithms	L4	CO1	**10**
		OR			
	b.	Evaluate the ethical concerns surrounding the application of machine learning	L2	CO1	**10**
9.	**a.**	Compare different machine learning algorithms with their strengths and weaknesses	L4	CO2	**10**
		OR			
	b.	Design a simple machine learning model for predicting house prices using a dataset	L4	CO2	**10**
10.	**a.**	Analyze the limitations of the Support Vector Machine algorithm	L4	CO2	**10**
		OR			
	b.	List various regression models, explaining where each would be best applied	L2	CO2	**10**
11.	**a.**	Discuss how backpropagation optimizes the learning process in machine learning	L2	CO3	**10**
		OR			
	b.	Discuss the significance of artificial neural networks in solving complex classification problems.	L2	CO3	**10**
12.	**a.**	Analyze how clustering techniques can be used in market segmentation	L4	CO4	**10**
		OR			
	b.	Design a clustering model to group customers based on purchasing behavior.	L6	CO4	**10**
13.	**a.**	Analyze the working of the Apriori algorithm and discuss its advantages and limitations in association rule mining	L4	CO5	**10**
		OR			
	b.	Compare various association rule mining algorithms and their applications.	L4	CO5	**10**